

# Minimum Wage Many-Analyst Experiment

## Executive Summary and Project Details

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# 1 Executive summary

This project tests whether **prompted literature priors** causally change what autonomous AI agents report about the employment effects of the minimum wage. Agents analyze the **same** state-year panel with the **same** task instructions; only the literature context (if any) is randomized.

**Design.** Five arms  $\times$  50 Cursor agents = **250** completed analyses. Arms: no prime; null-effect short; null-effect with citations; negative-effect short; negative-effect with citations. Agents choose outcome, sample window, treatment definition, covariates, and estimator under a shared DiD methodology guide.

**Headline results (Second Step run).**

| Finding                       | Result                                                                              |
|-------------------------------|-------------------------------------------------------------------------------------|
| Completion                    | <b>250/250</b> agents returned usable <b>results.csv</b>                            |
| Between-arm ATT differences   | <b>Not significant</b> (Fisher omnibus $p = 0.4101$ ; Kruskal-Wallis $p = 0.3063$ ) |
| Outcome choice                | ~ <b>99.6%</b> studied teen employment-to-population                                |
| Estimator choice              | ~ <b>98.8%</b> Callaway-Sant’Anna family; <b>4%</b> <b>TWFE</b> in any arm          |
| Treatment definition          | <b>249/250</b> <b>binary</b> ; <b>0%</b> <b>continuous</b> MW treatment             |
| Short vs citation-rich primes | <b>No meaningful ATT gap</b> on planned contrasts                                   |

**Bottom line.** With Cursor **composer-2** agents and Wave-2-style estimator discretion (TWFE allowed), **literature primes did not shift reported ATTs** in a detectable way. Behavior resembles Cunningham’s **constrained Wave 1** (CS-heavy, overlapping distributions) more than his **Wave 2** finding that negatively primed Claude Code agents moved toward TWFE, continuous treatment, and more negative estimates.

**Implication.** Primed priors alone are not enough: **platform, model, and instruction packaging** appear to mediate whether discretion produces estimator-channel divergence.

## 2 1. Purpose and motivation

Many-analyst studies show that researchers facing identical data can report different causal estimates through **specification choices**. Borjas & Breznau (2026, *Science Advances*) linked immigration researchers’ priors to their estimates. Scott Cunningham’s design turns that association into an **experiment** by randomizing the prior via a literature prime while holding data and task fixed ([Substack write-up](#)).

This repository is a **Cursor SDK replication / extension** of Cunningham’s **mw-analyst-experiment** design (originally Claude Code CLI).

**Research question:** Do literature primes about minimum-wage employment effects causally shift autonomous agents’ reported average treatment effects (ATTs)?

## 3 2. Experimental design

### 3.1 Arms

| Arm ID      | Label | Literature file |
|-------------|-------|-----------------|
| <b>none</b> | None  | ( <i>none</i> ) |

| Arm ID                      | Label          | Literature file                      |
|-----------------------------|----------------|--------------------------------------|
| <code>null_short</code>     | Null Short     | <code>PRIME_NULL_SHORT.md</code>     |
| <code>null_cites</code>     | Null Cites     | <code>PRIME_NULL_CITES.md</code>     |
| <code>negative_short</code> | Negative Short | <code>PRIME_NEGATIVE_SHORT.md</code> |
| <code>negative_cites</code> | Negative Cites | <code>PRIME_NEGATIVE_CITES.md</code> |

All agents receive: `agent_panel_essential.csv`, `DATA_DICTIONARY.md`, `DID_METHODODOLOGY.md`, and `INSTRUCTIONS_SHARED.md`. Primed arms also receive one literature context file.

### 3.2 What agents decide

- Employment outcome (CPS subgroups, QCEW industries, etc.)
- Treatment definition (binary staggered vs continuous)
- Sample years / states
- Covariates and comparison group
- Estimator (Callaway–Sant’Anna, continuous DiD, TWFE, etc.)

### 3.3 Deliverables per agent

1. `results.csv` — one row with design choices and ATT / SE / CI
2. `llms.txt` — structured narrative of the analysis

### 3.4 Data

Pre-aggregated **state** × **year** panel (51 units, 1990–2022): effective minimum wages, CPS employment rates for multiple groups, and selected QCEW industry series.

### 3.5 Scale and waves

| Wave                     | Location                     | Scale                    | Notes                                 |
|--------------------------|------------------------------|--------------------------|---------------------------------------|
| First                    | Frozen in <b>First Step/</b> | 150 agents (30×5)        | Earlier protocol;<br>TWFE discouraged |
| Second (current results) | <b>Second Step/results/</b>  | <b>250 agents (50×5)</b> | TWFE allowed in methodology           |

## 4 3. Technical pipeline

| Component                                                            | Role                                                   |
|----------------------------------------------------------------------|--------------------------------------------------------|
| <code>@cursor/sdk + launch_experiment_cursor.mjs</code>              | Local agents in isolated temp workspaces               |
| <code>run_full_experiment.sh / run_pilot_one_per_condition.sh</code> | Full (250) and pilot (5) launchers                     |
| <code>collect_results.sh</code>                                      | Aggregate outputs → <b>Second Step/results/</b>        |
| <code>analyze_results.R</code>                                       | Omnibus tests, plots                                   |
| <code>analyze_second_step_detail.R</code>                            | Extended tables/figures → <b>Second Step/analysis/</b> |

Default model: **composer-2**. Agents are sandboxed with **settingSources**: [] so user/project rules outside the workspace are not loaded.

## 5 4. Key results

### 5.1 Arm-level ATT summary

| Arm            | N  | Mean<br>ATT | SD     | Median  | % neg. | % teen | %<br>TWFE | Mean span |
|----------------|----|-------------|--------|---------|--------|--------|-----------|-----------|
| none           | 50 | -0.0050     | 0.0165 | -0.0091 | 78.0   | 100.0  | 0.0       | 26.5      |
| null_short     | 50 | -0.0064     | 0.0143 | -0.0080 | 80.0   | 100.0  | 2.0       | 25.7      |
| null_cites     | 50 | -0.0002     | 0.0209 | -0.0059 | 70.0   | 100.0  | 0.0       | 23.3      |
| negative_short | 50 | -0.0045     | 0.0137 | -0.0082 | 74.0   | 98.0   | 4.0       | 25.5      |
| negative_cites | 50 | -0.0053     | 0.0187 | -0.0098 | 76.0   | 100.0  | 0.0       | 27.4      |

### 5.2 Inference

- **Fisher omnibus** (between-arm sum of squares, 10,000 permutations): observed = 0.0011,  $p = 0.4101$ .
- **Kruskal–Wallis** across five arms:  $\chi^2 = 4.8195$ ,  $p = 0.3063$ .
- Planned contrasts (null – negative, short and cites): neither rejects at conventional levels.

**Interpretation:** condition labels are statistically exchangeable with respect to reported ATT in this run.

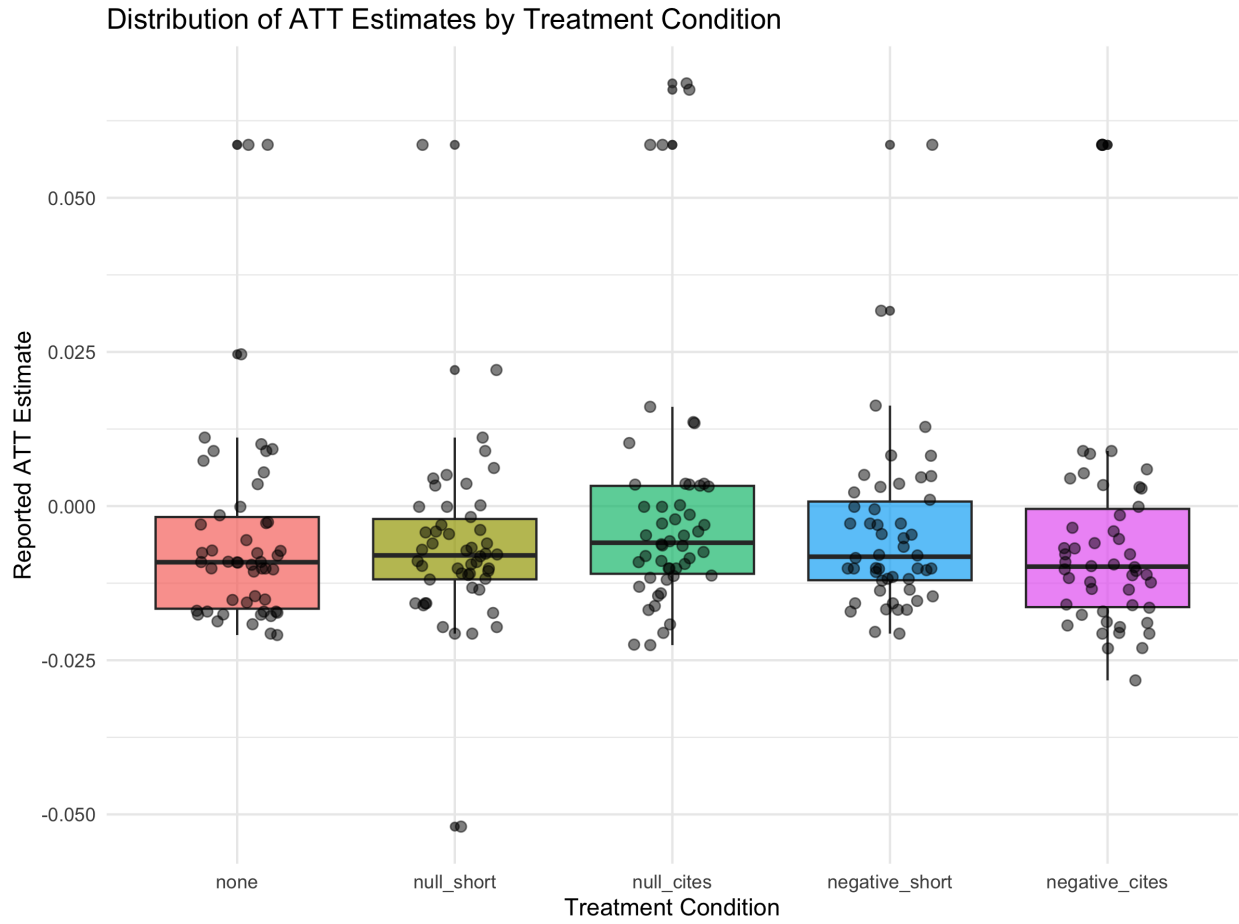


Figure 1: Distribution of reported ATT estimates by treatment arm ( $N = 250$ ).

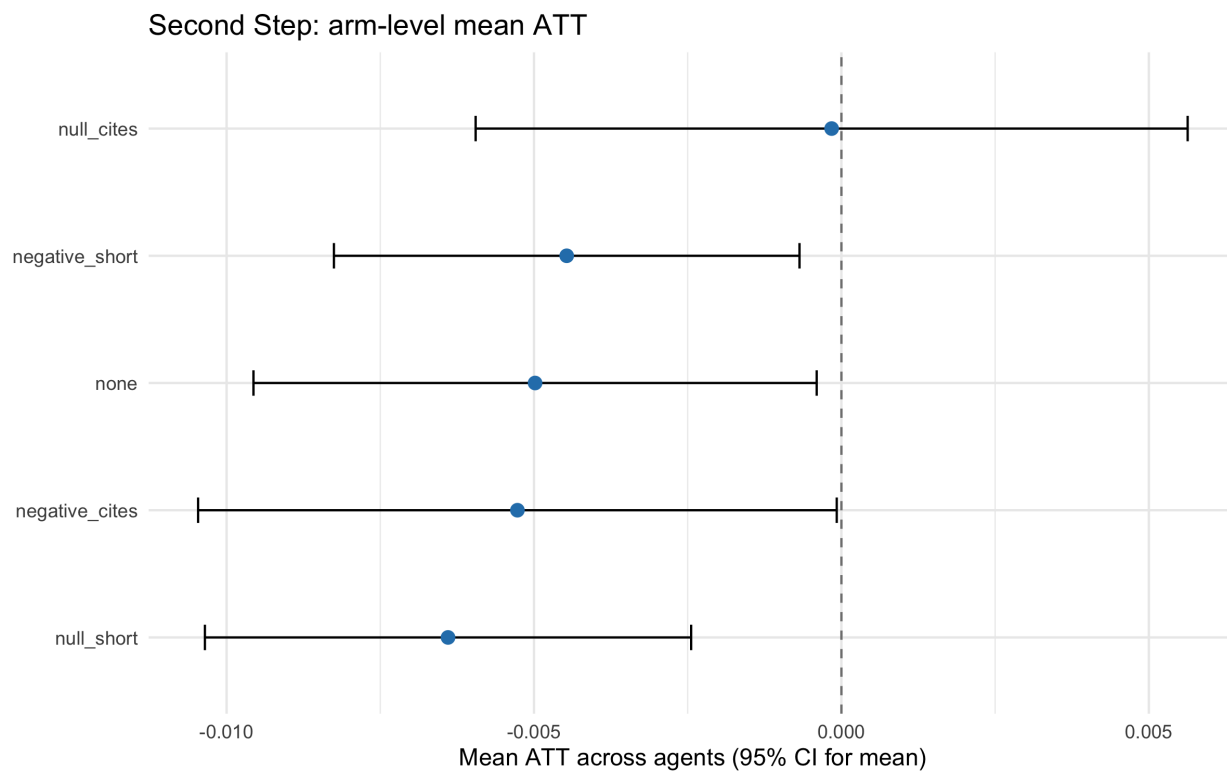


Figure 2: Arm-level mean ATT with approximate 95% intervals.

### 5.3 Specification patterns

- **Outcome:** nearly universal teen EMP/POP (`emp_pop_ratio_teen`).
- **Estimator:** Callaway–Sant’Anna dominates; TWFE rare and not concentrated in negative arms.
- **Treatment:** almost entirely binary staggered MW changes.
- **Covariates:** nontrivial share use none (~42–66% by arm); not clearly sorted by prime.
- **Event studies:** produced by 98% of agents per arm.

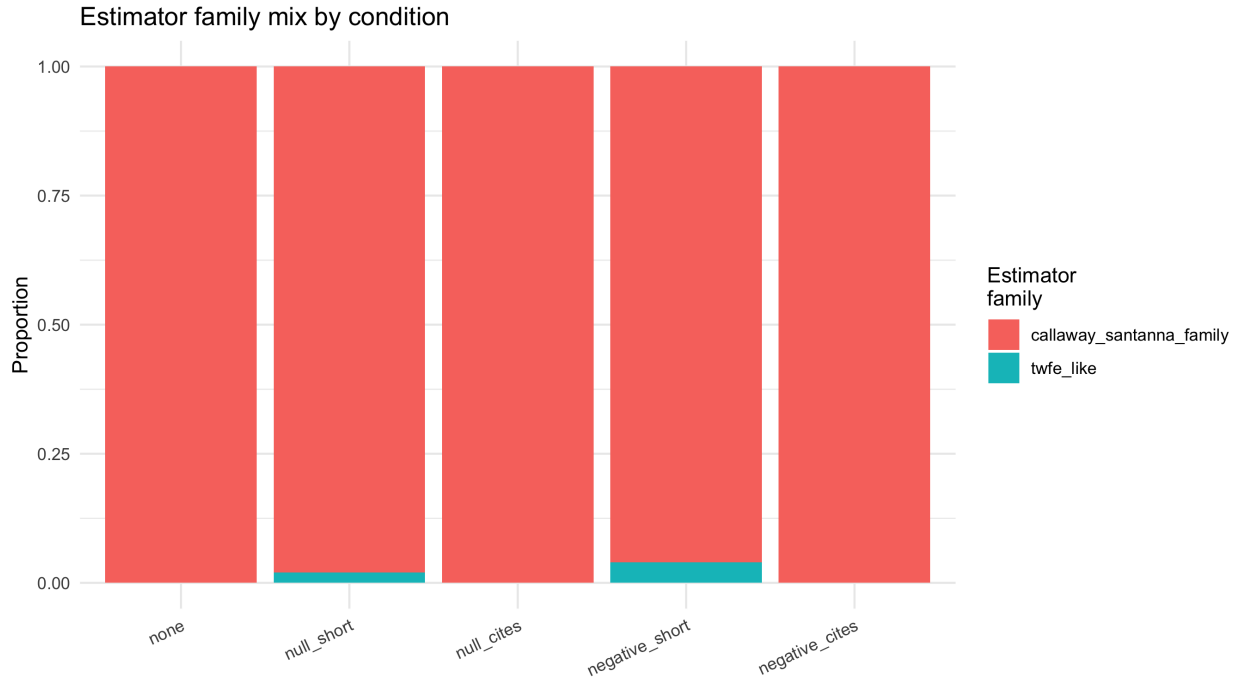


Figure 3: Estimator family composition by arm.

## 6 5. Comparison to Cunningham (2026)

| Dimension      | Cunningham (Substack)                                       | This Cursor run                                           |
|----------------|-------------------------------------------------------------|-----------------------------------------------------------|
| Platform       | Claude Code CLI                                             | Cursor SDK ( <code>composer-2</code> )                    |
| Arms           | 3                                                           | 5 (short vs cites splits + pure none)                     |
| Wave 1 pattern | Similar ATTs under CS-only                                  | <b>Replicates in spirit</b> (~99% CS; Fisher $p = 0.41$ ) |
| Wave 2 pattern | Negative arm → more TWFE, continuous Tx, more negative ATTs | <b>Does not replicate</b> (4% TWFE; 0% continuous)        |
| Data           | Same essential panel concept                                | Pre-aggregated CSV (no microdata rebuild)                 |

Most plausible explanation for the missing Wave 2 shift: **model/platform defaults** toward Callaway–Sant’Anna despite TWFE permission, possibly interacting with prime packaging and stricter Cursor sand-boxing. A fuller technical discussion is in `reports/cursor_replication_report.qmd`.

## 7 6. Repository map

```

First Step/          # Frozen first-wave materials
Second Step/
  results/           # 250-agent collected outputs + core plots

```

```

analysis/          # Extended summaries and figures
data/agent_panel_essential.csv
conditions.tsv
INSTRUCTIONS_SHARED.md / DID_METHODODOLOGY.md / PRIME_*.md
scripts/          # launch, collect, analyze
PROJECT_LOG.md    # Implementation history and troubleshooting
reports/          # This summary + full replication report

```

Reproduce analysis:

```

Rscript scripts/analyze_results.R "Second Step/results"
Rscript scripts/analyze_second_step_detail.R

```

---

## 8 7. Limitations and next steps

**Limitations.** Single model (`composer-2`); no rhetoric/confidence scoring pipeline; one permissive-estimator wave on Cursor (no within-platform CS-only vs TWFE-allowed contrast); prime texts differ from Cunningham’s exact wording.

**Testable next steps.**

1. Match Cunningham’s three-arm primes and Wave 2 methodology text verbatim.
  2. Re-run with a Claude model via Cursor SDK (if available), holding instructions fixed.
  3. Inspect agent logs for abandoned TWFE trials (mechanism check).
  4. Optionally relax sandbox (`settingSources`) to test ambient-memory hypotheses.
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## 9 References

- Cunningham, S. (2026). “Claude Code 45: AI Agents and the Minimum Wage.” *Scott’s Mixtape Substack*. <https://causalinf.substack.com/p/claude-code-45-ai-agents-and-the>
- Borjas, G. J., & Breznau, N. (2026). Many-analyst study of immigration. *Science Advances*.
- Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., & Sant’Anna, P. H. C. (2025). Difference-in-Differences Designs: A Practitioner’s Guide.
- Sant’Anna, P. H. C. (2025). *My Claude Code Setup*. Emory University. <https://psantanna.com/claude-code-my-workflow/>

## 10 Acknowledgments

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Experimental design and original many-analyst concept: Scott Cunningham, Baylor University.